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Cradle Drive Solenoid

christopher hardiman (christopher c hardiman), 04/24/2002 - 03:39 pm



PROPRIETARY WARNING

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Communique Number

MQUE-59HS7R

Equipment

LightSpeed Plus and LightSpeed Ultra

Problem

A vendor manufacturing issue has been identified by Tom Neitzel and Bob Leitingner with the LightSpeed Plus and Ultra cradle drive assembly. At issue is the fact that the vendor is setting the Cradle Release Solenoid so far back in its adjustable mount that it is unable to draw the plunger into the solenoid body thereby engaging the motor to the rollers. After a new install, the system gets Cradle Velocity Errors. The gantry display will show that the cradle is latched; however, it will free float if pushed or pulled. Tightening the slip clutch does not help and the problem is intermittent. Some FEs have manipulated the shimming at the rear of the cradle to try to resolve the problem to no avail.

Solution

Please refer to directions 2153315-100 rev 8, page 510, item 7 or 2296435-100, rev 3, page 494, item 7 for details on how to correct the problem. More details are contained in the attached file.

You can download any of the files below by dragging them to your Windows desktop.
 To open a file for reading, double-click it.



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Revision

04/24/2002 03:39 PM

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Cradle Release Solenoid Issue at Install – LightSpeed Plus and LightSpeed Ultra Systems

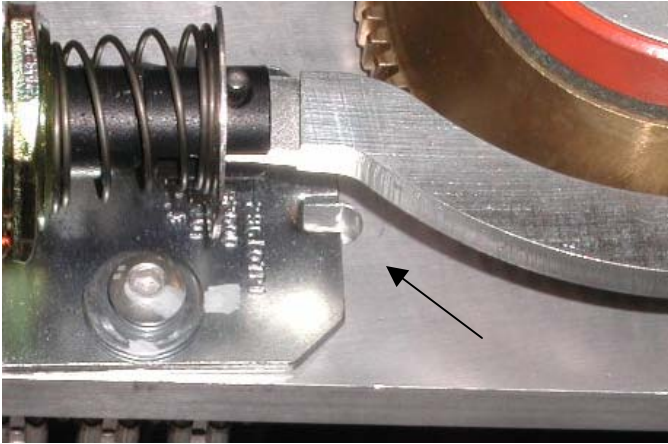


Figure 1. Misadjusted Solenoid

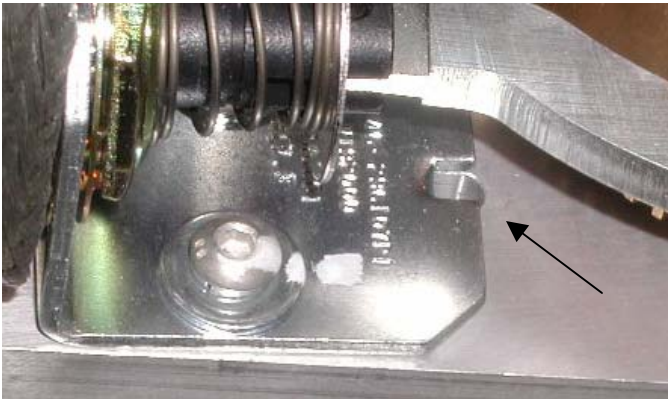


Figure 2. Properly Adjusted Solenoid

As seen in Figure 1, the solenoid is screwed and locked in the furthest back position. In a few cases, we have found it as far back as it can go. This can cause intermittent problems with the plunger not engaging resulting in Cradle Velocity Errors.

If the adjustment is done properly, it should look more like Figure 2.

Note: Make sure that it's not too far forward so that the gears don't mesh completely. If they don't, it can cause jerky table movement resulting in position or velocity errors.

The proper way to adjust the solenoid per the service manual procedure 2296435-100 section 5.4.1, step 7 follows:

"When the solenoid is energized, the gear rack is engaged in the clutch gear and allows the cradle to be driven. The engagement of the rack in the gear must not have any backlash, nor can the solenoid plunger be excessively extended out of the solenoid body. When correctly set, the solenoid plunger will be within 0.010" of bottoming-out in the body, when there is no backlash at the rack/gear interface.

Adjust the solenoid bracket so that the plunger is bottomed when the solenoid is energized, and then move the bracket forward (toward the gantry) until there is no backlash between the rack and gear, as checked at four, 90 degree apart, positions on the gear. The solenoid should be energized so that all the looseness is removed from the linkage; if energizing is not possible, be sure to push on the plunger itself (not the pin or link) when checking the adjustment."

Note: Refer to Figure 3. Another potential source of failure in this area, that would have the same symptoms, involves the distance between the gear rack and the clutch gear. This distance should be adjusted to .025 +/- .002 mm.

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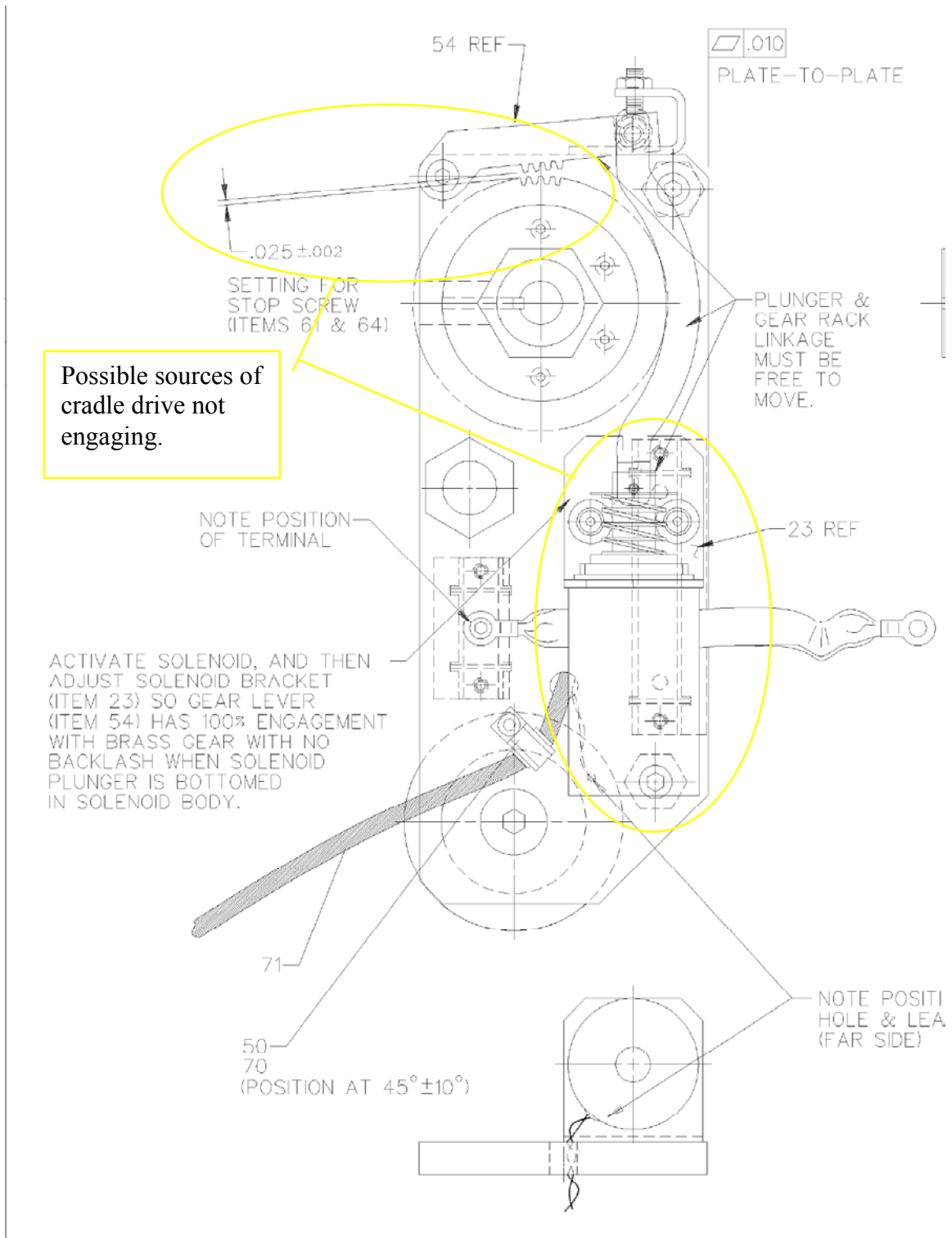


Figure 3. Possible Sources of Cradle Drive Failing to Engage